

A Survey DDoS Attack in IoT Cloud Platform

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Abstract—The Internet of Things (IoT) and cloud computing are the key components in the industrial revolution. The IoT platform provides provisioning, administration, and automation of smart items inside a specific IoT architecture, enabling the goal of connecting physical objects to the Internet so they may collaborate, communicate, and act intelligently without the need for human involvement. On the other hand, cloud computing has brought consumer and business applications online, revitalizing IT services and saving businesses money by eliminating the need to build and maintain on-site IT infrastructure for data storage and application operations. The long-term cost-effectiveness of cloud-based solutions includes enhanced security, corporate data mobility, more employee cooperation, and more sophisticated disaster recovery plans. The majority of cyber physical systems are currently hosted on IoT cloud platforms due to their advantages. However, security issues of the platform have to be addressed. Distributed Denial of Service attacks (DDoS) are among the most common. This paper studies the importance of IoT cloud architecture, discusses the defensive measures against DDoS attacks, and suggests a high-level conceptual IoT architecture to defend DDoS attacks while addressing future research challenges.

Index Terms—IoT, Cloud, DDoS, IoT cloud architecture, cyber physical system.

I. INTRODUCTION

The technologies are increasing in a rapid way. The advent of the Internet of Things (IoT) enables the easy transfer of data over a connected network from any physical object without human intervention [1]. IoT has a three-layer architecture, namely the perception layer, network layer, and application layer. All physical objects, sensors, and actuators collect information in the perception layer. The network layer is the communication layer, where all information is transferred securely. All application processes are managed by the application layer. Network gateways are added for connecting heterogeneous networks and cloud environments. The emergence of cloud computing delivers more reliability, flexibility, and efficiency, which improves performance and innovation and translate the organization to the industrial market in a faster time by incorporating cognitive learning using machine learning (ML) and artificial intelligence (AI) in their use cases [2]. Thus, cloud computing brings a drastic change in industries by translating everything as a service, whereas the IoT brings a radical change in everyday life by connecting everything to

the cyber world. Now, the world is becoming a part of the IoT-cloud network. The services offered in the IoT-cloud platform have capabilities to analyze critical business, minimize risk, and reduce operational and developmental costs by providing end-to-end agile solutions [3]. The amalgamation of these technologies brings forth advantages not only in the industrial field but also in our everyday lives.

The IoT devices amplify the effect of DDoS attacks owing to their vulnerability to security concerns [4]. Since IoT devices lack built-in security due to their limited hardware and computational capabilities, the attackers can compromise IoT devices and can use them as botnets for DDoS attacks. These security challenges arise regardless of the added value that the IoT-cloud platform brings to business applications and consumers. Distributed Denial of Service (DDoS) is one of the most successful attacks to date. Furthermore, the risk of attacks is increased by the dispersion of IoT devices throughout the network and the data flow from IoT devices to the cloud network via the Internet.

The recent statistics published by A10 show that the DDoS attack has hovered around 15.1 million for the past few years [5]. The primary cause of the surge in cyberattacks was COVID-19, which caused everything to move online. In February 2020, a 2.3 gigabytes per second DDoS attack is directed at AWS. On October 6, 2020, a UDP amplification attack of 167 Mpps (millions of packets per second), four times more powerful than the Mirai botnet attack, was launched against Google. AWS, Google, and Azure experienced a record-breaking DDoS attack: HPPT/2 rapid reset hits multiple targets in 2023, which leads to resource exhaustion. Figure 1 displays the Cloudflare report of HTTP DDoS attacks by year [6]. This attack is 201 million requests per second (rps), which is over eight times greater compared to our prior record of 26 million rps from 2022.

The Cyber Report by Vercara on DDoS attacks by industry in 2023 is shown in Figure 2 [7]. The data indicates industrial sectors targeted by DDoS attackers in 2023. The top three industries targeted by DDoS attacks in 2023 were communication service providers, financial services, and software/web services.

Due to the increase of different variants of DDoS attacks and the different vulnerabilities of IoT devices, the security be-

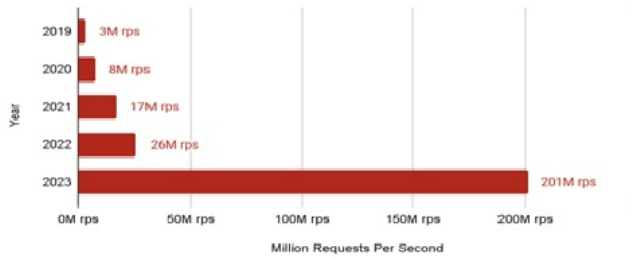


Fig. 1. Cloudflare report of largest HTTP DDoS attacks by year [6].

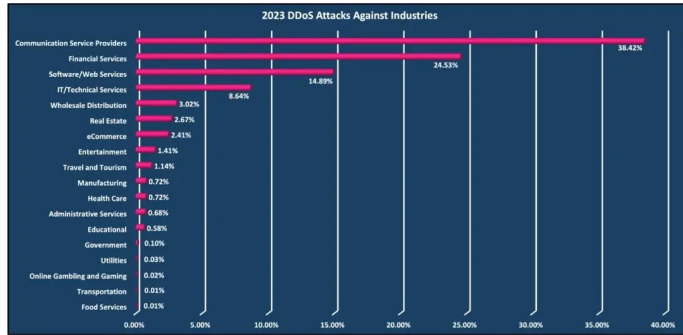


Fig. 2. The Cyber Report by Vercara on DDoS attacks by industry in the year of 2023 [7].

come a major concern in this IoT-cloud era. Many researchers have been working against DDoS attack using AI or ML algorithms [8]. Yet, the recent attacks show the evidence of DDoS is still there. This article discusses the various measures for detecting and mitigating DDoS attacks, the significance of IoT cloud architecture in defending DDoS attack and conclude with future challenges in deploying it.

II. AN OVERVIEW OF IoT CLOUD ARCHITECTURE

A. Significance of IoT Cloud Architecture

An IoT cloud platform is where cloud computing and IoT tech stacks are combined to provide new value for commercial and consumer applications. In fact, the increasing use of cloud services has accelerated the growth of IoT in many industries. The primary factors for convergence of cloud and IoT technologies are:

- IoT often have limited processing and energy resources. This prevent complicated data processing on IoT devices. IoT devices will be able to process data on-site by utilizing cloud computing capabilities [3].
- Although it might be costly to construct an IoT infrastructure on-site, there are financial advantages to keeping your data in the cloud [9].

However, there may be security issues with IoT cloud architecture at the IoT device level, networking and communication level. So, IoT cloud security is crucial for it protect the gadgets and sensitive data from the internet threats.

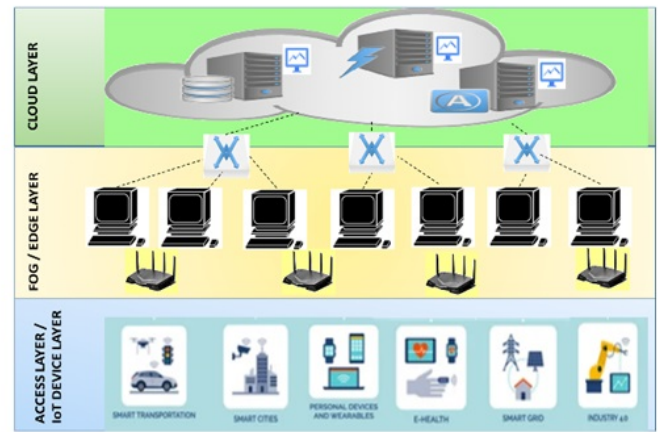


Fig. 3. IoT Cloud Architecture

B. Functionalities and Vulnerabilities of IoT Cloud Architecture

The functionalities depend on the three layers of IoT cloud architecture shown in Figure 3. The first layer is the IoT device layer or sensing layer, where data is collected from the physical environment through sensors, actuators, or embedded systems. The second layer is the edge layer, which contains a network of IoT gateways and edge nodes. It can be wired or wireless. The main functionality of this layer includes routing, adoption, and fragmentation. Finally, user data and requests are transformed into demands for services by the application layer and forwarded to the lower tiers of the communication system.

The vulnerabilities of IoT cloud architecture include insecure network connections, software and hardware flaws, inadequate authentication and authorization, insufficient transport encryption, unsafe network services, a lack of security configuration, and physical security threats [10].

C. DDoS Attacks on IoT Cloud Architecture

Based on the layers of IoT Cloud Architecture, DDoS Attack are classified into three.

1) *Perception Layer Attack*: IoT devices are distributed around the globe, where hackers may easily and rapidly obtain access, leaving them open to a variety of security threats [11] [12]. MAC flooding and physical layer flooding are the two main categories under which cyberattacks fall in this tier.

2) *Network Layer Attack*: The purpose of a DDoS attack is to overload the IoT network's capacity and reduce its ability to reply to legitimate requests, thus interfering with its normal operation. An attacker may target individual devices, gateways, or even the network infrastructure itself in an attempt to exploit vulnerabilities and compromise network availability. The two main types of attacks on this network layer are ICMP flood attacks and IP-based attacks. It includes ping of death attacks, land attacks, teardrop attacks, ICMP spoofing, smurf attacks, ICMP ping flood attacks, TCP-SYN attacks, UDP

attacks, ACK-PUSH attacks, RST-FIN attacks, Sybil attacks, and Sinkhole attacks [4].

3) *Cloud Layer Attack*: Through the application of strategies such as HTTP (GET/POST) calls and various queries directed at system software—including Windows, Apache, and OpenBSD—malicious actors targeting the application layer endeavor to compromise the integrity of IoT network infrastructure. As the application layer leverages cloud infrastructures and services, three distinct categories of DDoS attacks have been delineated based on the associated cloud components [13].

- 1) **Cyberattacks Targeting Cloud Infrastructure**: This refers to attacks on hypervisors, virtual machines, and cloud schedulers. It includes attacks like flooding attacks, carpet bombing, smurf and fraggle, multi-vector attacks, and cloud internal denial of service (CIDoS).
- 2) **Cyberattacks Against Cloud Based Services**: This refers to attacks against cloud web services and software as a service (SAAS). It consists of the following: Slowloris, RUDY, SIP Flood, NTP Amplification, DNS Amplification, HTTP Flood, and Oversized Encryption Attack.
- 3) **Cyberattacks Targeting Cloud Customers**: Cyberattacks directed against cloud customers. The attacks known as Economic Denial of Sustainability (EDoS) targets the customer's financial resources by charging them for unauthorized resource consumption.

These attacks on the architecture of the IoT Cloud demonstrate the need to establish a DDoS protection design. Recent investigations have utilized artificial intelligence and machine learning methodologies for the detection and mitigation of DDoS attacks. A thorough literature review primarily concentrated on critically evaluating the current techniques designed to safeguard IoT cloud platforms from distributed and large-scale traffic-based attacks.

III. STATE OF ART

A. ML Based Approaches for DDoS Attacks

Aysa *et al.* [14] presented a lightweight detection model using correlation based feature selection in 2020. The authors created a uniform dataset from three different Wi-Fi IoT camera devices. A good accuracy is achieved from the synthetic dataset used. In this study, machine learning algorithms such as neural networks (NN), linear support vector machines (LSVM), J-48, and random forests (RF) were employed. Significantly, the integration of RF with decision tree (DT) methodologies achieved the highest accuracy.

A sequential attack detection framework was used in the IoT-botnet attack detection method presented by Soe *et al.* [15]. This technique obtained 99% accuracy in the N-BaIoT dataset using DT, Naive Bayes (NB), J48, and Artificial Neural Network (ANN) classifier. The “Model Builder” and the “Attack Detector” are the two stages of this system. The network is classified in two group: legitimate and attack. Nevertheless, our method was unable to differentiate between several DDoS attacks classifications.

A unique framework model called CorrAUC was presented by Shafiq *et al.* [16] with a useful feature selection method. This framework was created specifically to spot malicious traffic and abnormalities in IoT networks. Pearson correlation, AUC and class label are used for feature selection and ML based model is built using DT, SVM, NB, and RF classifiers employing the Bot-IoT dataset. According to their claims, compared to other approaches, the DT-based model produced classification accuracy that was around 99% higher. This strategy has not succeeded in offering a more effective defense against widespread DDoS attacks; instead, its primary objective is to shield the victim from DoS attacks.

A DDoS detection technique based on IoT traffic was published by Doshi *et al.* [17] for identifying DDoS attacks generated by IoT. Here, the network feature is classified into stateless features including protocol and size in packet header and stateful features including bandwidth, source IP, destination IP etc. Using these network features, construct a detection model using several ML methods such as DT, LSVM, RF, K-nearest neighbors (KNN), and NN, which efficiently classifies the incoming traffic with 99% accuracy. But since this system was built using an uneven dataset, its output might be skewed in favor of the dominant class. Nevertheless, strong botnets are difficult to identify through this ML model.

Churcher *et al.* [18] presented a detection strategy in a realistic Bot-IoT dataset using a variety of ML algorithms, including SVM, RF, DT, KNN, NB, ANN, and LR. KNN-based method has good performance when they tested using 1.5 million records with 99% classification accuracy. Since the system is built using imbalanced data, it can give false findings in practical situations. Furthermore, this method is unable to discern the traffic flows from DDoS attacks from flash events.

In order to provide a ML-based technique for identifying DDoS-infected traffic in consumer IoT (CIoT), Gupta *et al.* [19] examined characteristics of IoT networks. A dataset is produced by collecting normal and malicious traffic through IoT network. Six machine learning models—the KNN, NB, SVM, DT, LR, and RF algorithms—were employed for identification based on the acquired dataset. Additionally, the local router screened out anomalous traffic as a mitigating measure. RF classifier delivers higher performance in attack detection with low false alarms.

Shukla *et al.* [20] presented a DDoS detection strategy based on IoT traffic that used an embedded feature reduction technique to achieve a high detection rate. The multiclass machine learning technique is used to classify the incoming IoT traffic into 11 classes. Effective model classification with embedded feature reduction approach is executed in Bot-IoT and minimize the overfitting issues. The findings demonstrate that the suggested strategy offers a feature reduction rate of 84.4% and around 5.19% greater classification accuracy

B. DL Based Approaches for DDoS Attacks

To ensure the security of IoT networks, Larriva *et al.* [21] suggested a DL-based Intrusion Detection System (IDS) technique. They developed their models using three different

datasets: NSL-KDD, UGR16, and UNSW-NB15. On these datasets with predefined classifications, they used z-score, min-max, and several pre-processing techniques. To categorize network flaws, they used MLP classifier. For NSL-KDD, UNSW-NB15, and UGR16, respectively, the classification accuracy of this system is 99.7%, 99.2%, and 99.3%. Nevertheless, there is no comparison of the effectiveness of the suggested approach against different attack types. Additionally, this method was unable to discern DDoS attack packets from flash events.

To identify botnet intrusion in IoT networks, Popoola *et al.* [22] implemented a DL-based method. As this detection approach using a highly imbalanced Bot-IoT dataset, asymmetric network traffic data in the Bot-IoT dataset maintains equilibrium and the issues of overfitting or underfitting are minimized by employing the synthetic minority oversampling (SMOTE) technique. Next, they deployed the Deep Recurrent Neural Network (DRNN) model once the data had been normalized. It will take longer to gather all the information, and can subject to a large-scale attack, because no feature selection strategy was included in the detection approach's design.

Hezam *et al.* [23] suggested a combined method of Convolution Neural Network (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) models for detecting DDoS botnet attacks. CNN is responsible for preprocessing and feature optimization and BiLSTM is used to identify botnets in the network like Mirai and Bahlite botnets. To validate our approach, nine hacked IoT devices' attack traffic is included in a realistic N-BaIoT dataset. Research is being conducted on the performance of four DL-based models, which include RNN, LSTM-RNN, CNN, and BiLSTM-CNN and the BiLSTM-CNN model has been found to outperform other models in terms of performance. Although, the BiLSTM-CNN model detects botnet attacks, it is unable to counteract the sophisticated and often occurring cyberattacks of today.

A DL-based attack detection method was presented by Koroniotis *et al.* [24] utilizing LSTM, SVM, and RNN algorithms. Statistical approaches such as correlation coefficient and joint entropy were utilized to extract the top 10 characteristics from the synthetic dataset in order to construct these models to categorize attack and genuine packets. However, it is unable to differentiate between various DDoS attacks.

A technique based on deep neural networks (DNNs) was presented by Kim *et al.* [25] to safeguard networks against various security risks. This method was created with the KDD-1999 dataset. Stochastic gradient descent (SGD) optimization algorithm in conjunction with a ReLU activation function was utilized. The proposed model achieved a classification accuracy of 99% for network traffic. However, this approach was inadequate in safeguarding the system against large-scale DDoS attacks.

The IoT technological advancements like edge computing and fog computing allow extensible and continual monitoring solutions. Additionally, sophisticated algorithms like ML, DL and ensemble methods are able to comprehend complex

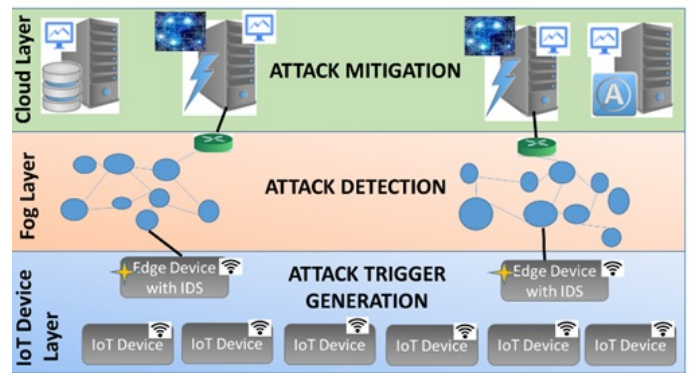


Fig. 4. Proposed Conceptual Model of IoT Cloud Architecture for DDoS attacks

patterns and identify hidden irregularities, resulting in more accurate and dependable detection outcomes. Network traffic may be examined locally, cutting latency and facilitating quicker reaction times, by directly implementing machine learning models on edge devices or fog nodes [26]. However, considerable computing resources and knowledge may be needed to execute cutting-edge machine learning algorithms in IoT-based networks.

Based on the current state of the art, it is evident that while many recent approaches achieve high accuracy, existing detection strategies and architectures are inadequate for identifying large-scale DDoS attacks and countering modern, sophisticated cyber threats with minimal detection latency. These methods struggle to distinguish between flash events and various categories of DDoS attacks. Furthermore, the available imbalanced and synthetic datasets fail to provide accurate detection results. These limitations represent significant gaps in the existing state-of-the-art methods and frameworks.

IV. HIGH LEVEL CONCEPTUAL MODEL OF PROPOSED METHOD

The IoT aims to make our lives more efficient and give us total control. Sensors on IoT devices allow for the autonomous collection of vast volumes of data. However, the low processing power and storage capacity of IoT devices highlights the cloud's computing capabilities. IoT device powering services and infrastructure are supplied from the cloud to form IoT cloud platform. The latency of a cloud-centered architecture impacts the flow of data between IoT gadgets and the cloud. The fog layer is introduced for this reason and more in order to assist in resolving challenges with cloud architecture by moving it closer to the edge of the network.

By considering the architectural benefits of the IoT cloud platform and the recent works related to DDoS attacks, a high level conceptual model is proposed as shown in the Figure 4. Thus, the proposed conceptual model is divided into three primary layers: the cloud layer, the fog layer, and the IoT device layer. Every IoT device collects data according to its application at its initial layer, which is the physical layer. It includes sensors, actuators, and edge devices

to communicate with the physical world. The edge devices embed with an integrated IDS in addition to the standard IoT design. Considering the shortcomings of the IoT device, the sensing layer incorporates a lightweight anomaly detection algorithm to generate alarms. Since anomaly detection has a higher chance of a false positive, the edge device can direct traffic for the fog layer to analyze the traffic and confirm the attack probability before going to the application without any latency. Thus, the fog layer consists of a cluster of machines where the data from the access layer is distributed and stored. It runs with a powerful detection algorithm for DDoS attack detection. By slicing network flows across a group of machines in the fog layer, the detection procedure can be carried out concurrently to safeguard the infected machine quickly. The cloud application layer counteracts the attacks once detection of attacks has been verified. The cloud layer implements the defenses and routes in-depth reports to the dashboard administrator. It also sends back feedback on attack footprints to the lower layer for future reference. By deploying cognitive learning with the AI model on the cloud layer, it improves the accuracy of DDoS attack detection and mitigation.

Thus, the methodology outline involves several key steps. The first step is identification of the challenges associated with current DDoS detection methods in IoT environments and to recognize the critical need for real-time, efficient, and precise detection of large-scale, sophisticated DDoS cyberattacks. The second step focuses on modeling the proposed system, taking into account the limited computational resources of IoT devices. This leads to the introduction of the cloud layer, which is responsible for large-scale data processing, advanced threat mitigation, and long-term data storage and analysis. The fog layer is then introduced as an intermediary between the cloud and IoT devices, designed to reduce latency and improve real-time decision-making by distributing the detection workload across multiple machines. Edge devices play a key role by quickly processing data locally, filtering out irrelevant traffic, and forwarding only pertinent information to the fog or cloud layers for deeper analysis.

The third step is to integrate AI and cognitive learning into the system, enabling it to learn from past attack patterns and continually enhance its ability to detect and mitigate future threats. Edge devices can be embedded with simple machine learning algorithms to perform initial data filtering and preliminary threat detection, offloading this task from the fog and cloud layers. The fourth step is to validate the model by testing it with various DDoS attack scenarios, ensuring its effectiveness in real-world conditions. The fifth step involves iterating and refining the model by implementing feedback loops, where detected attack patterns are returned to the system for future learning and improvement. The model's performance is continuously evaluated, with detection algorithms being updated and new features added to handle emerging attack techniques. The final step is to deploy and monitor the system in a live environment, where it will be regularly assessed, configurations adjusted as necessary, and the system will be kept up-to-date to defend against new and evolving

threats.

V. RESEARCH CHALLENGES AND CONCLUSIONS

A. Research Challenges

- 1) Currently, the IoT Cloud platform lacks a common architecture. Consequently, a distinct architectural framework is used to construct the defensive approach.
- 2) For real-time traffic, DDoS databases are unreliable.
- 3) Inability of detection strategy to find the zero day attacks.
- 4) The majority of the DDoS defense is focused on volumetric attacks, not flash crowd attacks.
- 5) Smart data management is not possible for IoT devices due to its distributed nature.
- 6) In order to preserve Quality of Service (QoS), defense strategies should take into account the networking flexibility and the balance of hardware and software utilized in IoT.
- 7) Limited research has been conducted on DDoS defense in the IoT cloud platform.

B. Conclusion

Cloud computing and IoT come from distinct realms. Because IoT and cloud are complimentary, integrating them is advantageous for both. The cloud's practically limitless resources offset the Internet of Things' resource limitations. Additionally, the IoT has the potential to create new real-world services and extend cloud services in a more distributed manner. Consequently, the IoT-cloud platform offers two benefits: firstly, it allows IoT providers to take advantage of cloud computing features like resource pooling, on-demand self-service, wide network coverage, measured service, and quick elasticity; secondly, it relieves the resource-constrained IoT technologies' high demands for data processing and storage.

In addition to reducing resource-constrained behavior and enhancing IoT technology system performance, the IoT cloud opens up new possibilities for developing and implementing IoT security solutions. Certain IoT devices have time-sensitive applications; therefore, the data that is gathered must be processed and evaluated promptly. Cloud-centralized design induces latency in communication between IoT devices and the cloud, as security applications are time-sensitive. This can be solved by introducing fog layer. This article examines several DDoS attack defense tactics and develops a suggested paradigm for the IoT cloud platform where fog computing holds the key to novel insights in networks.

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